Paula Luna-Velasco

Curriculum Vitae | pluna@us.es | Personal webpage | Seville, Spain

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Personal Details

Institutions	University of Seville (US), University of Granada (UGR)
Postal address	Edificio Celestino Mutis, 1st Floor, Campus de Reina Mercedes, Avenida Reina Mercedes s/n, 41012 Seville, Spain
Research interests	Analysis, Partial Differential Equations, Fluid Mechanics
Languages	Spanish (native), English (C1), French (C1), Italian (A2, currently studying B1)

Education

2023–2027	**PhD in Mathematics**, University of Seville (US & IMUS), Spain *Supervisors:* Francisco Gancedo García and Eduardo García-Juárez *Thesis:* Fluid Interfaces: Well-Posedness, Regularity and Instabilities
2022–2023	**Master's Degree in Advanced Mathematics**, Complutense University of Madrid (UCM), Spain *Master's thesis:* A study of elliptic regularity through the smoothing properties of evolution semigroups. *Supervisor:* Aníbal Rodríguez Bernal
2018–2022	**Bachelor's Degree in Mathematics**, University of Seville (US), Spain *Bachelor's thesis:* Derivation and solution of the linear water wave equations *Supervisor:* Francisco Gancedo García
2020–2021	**Erasmus exchange (BSc in Mathematics)**, Sorbonne University, Paris, France

Publications

[2] *Mixing zone dynamics for the Saffman–Taylor instability with viscosity jump*, with F. Mengual. Preprint (2026). arXiv:2610.34504.

[1] *On 2D Navier–Stokes free boundary: nonnegative density and small viscosity contrast*, with F. Gancedo and E. García-Juárez. Submitted (2025). arXiv:2507.09333.

Research Fellowships

2023–2027	**Predoctoral Fellowship FPI 2022 – María de Maeztu Excellence Unit** (PRE2022-105212), Government of Spain Institute of Mathematics of the University of Granada (IMAG)
2022–2023	**Collaboration Scholarship**, Government of Spain Complutense University of Madrid (UCM)

Research Stays

Feb–Mar 2027	**Carnegie Mellon University (CMU)**, Pittsburgh, USA *(upcoming)* *Host:* Ian Tice
24 Apr–4 Jul 2026	**Université Paris-Est Créteil (UPEC)**, Paris, France. Funded by the FPI fellowship *Host:* Raphaël Danchin
21 Oct–9 Nov 2024	**Institute for Advanced Study (IAS)**, Princeton, USA *Host:* Francisco Gancedo

Talks and Posters

Jan 2027	**Invited talk**, RSME Young Researchers Congress, Zaragoza, Spain *(upcoming)*
12 Nov 2026	**Seminar**, Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain *(upcoming)*
7 Jul 2026	**Invited talk**, XXIX CEDYA / XIX CMA, Valencia, Spain
29 Jun 2026	**Poster**, Workshop: Idealised mathematical models for geophysical flows, Institut Henri Poincaré (IHP), Paris, France
1 Jun 2026	**Seminar**, Differential Equations Seminar, Dalian University of Technology (DUT), China

24 Mar 2026	**Seminar**, Graduate Student Analysis Seminar (GSAS), Purdue University, USA (online)
24 Feb 2026	**Seminar**, PHD Seminar, IMUS, University of Seville, Spain
19 Nov 2025	**Contributed talk**, 7th BYMAT Conference, IMUS, Seville, Spain
5 Jun 2025	**Seminar**, Young Researchers Seminar, IMAG, Granada, Spain

Participation in Funded Projects

2026–2029	**Fluid Interfaces: Structures, Singularities, Asymptotics** National project PID2025-172153NB-I00, AEI (Spain). PI: E. García-Juárez (US).
2026–2029	**Deterministic and stochastic PDEs in biology and fluid mechanics** National project PID2025-171734NA-I00, AEI (Spain) PIs: D. Alonso-Orán and M. Magliocca (University of La Laguna)
2025–2028	**Stability in incompressible fluid mechanics** National project PID2024-158418NB-I00, AEI (Spain). PI: Á. Castro (ICMAT–CSIC).
2023–2027	**Global dynamics, geometry, and regularity of incompressible fluids** National project PID2022-140494NA-I00, AEI (Spain) PIs: E. García-Juárez and F. Torres de Lizaur (US)
2023–2024	**Incompressible fluid interface** Project FIUS23/0207, Research Foundation of the University of Seville (FIUS) PI: F. Gancedo (US)

Teaching Experience

Oct–Dec 2026	**Mathematics (6 ECTS)** – BSc in Chemistry, 1st year, US
Sep 2025–Jan 2026	**Mathematics (6 ECTS)** – BSc in Chemistry, 1st year, US
Mar–May 2025	**Elements of Numerical Calculus (2 ECTS)** – BSc in Statistics, 2nd year, US
Sep–Nov 2024	**Mathematics (4 ECTS)** – BSc in Chemistry, 1st year, US
Jan–Jun 2023	**Numerical Analysis (6 ECTS)** – Engineering programme (exchange students), Comillas Pontifical University – ICAI, Madrid

Training in Teaching and Communication

Feb 2025	**Online session**: *Experiencias docentes de profesores juniors* (2 h), UGR
Jan 2025	**Science communication course**: *¡Comunicad Insensatos!* (20 h), DaSCI Institute, University of Granada. Storytelling, presentation and poster design, oral presentation

Conferences, Workshops and Schools Attended

May 2026	**Workshop: Analysis of PDEs in Dalian**, DUT, Dalian, China †
May 2026	**Workshop 2: Instabilities and transitions in geophysical flows**, IHP, Paris, France
Oct 2025	**End of CRISIS: a mathematical school and conference on fluid dynamics**, CIRM, Marseille, France †
Jul 2025	**Course: Bifurcation theory and overdetermined problems** *Lecturer:* D. Ruiz (10 h) – IMUS, Seville, Spain
Jul 2025	**NSF–UEFISCDI Summer School**, IMAR, Bucharest, Romania †
Jun 2025	**Conference on Advances in Mathematical Fluid Dynamics**, Peking University (PKU), Beijing, China †
Jun 2025	**Minicourse: Long-time dynamics of the incompressible Euler equations** *Lecturer:* I.-J. Jeong (6 h) – PKU, Beijing, China †
May 2025	**Cabo de Gata PDE Days**, Almería, Spain
May 2025	**BIRS–IMAG Workshop: Singularities and Asymptotic Patterns in Fluids and Evolutionary PDEs (25w5330)**, IMAG, Granada, Spain †
Feb 2025	**Fluid Mechanics at the ICMAT**, ICMAT, Madrid, Spain
Dec 2024	**Course: Renormalized solutions of elliptic equations with L^1 data** *Lecturer:* F. Murat (5 h) – IMUS, Seville, Spain
Oct 2024	**PDEs in Tenerife: From Theory to Applications**, University of La Laguna, Tenerife, Spain
Jul 2024	**Alhambra PDE Days (satellite event of the 9th ECM)**, IMAG, Granada, Spain

May 2024	**Summer School: (In)-Stability Phenomena in Fluid Mechanics**, CY Cergy Paris University, France †
Apr 2024	**Bilbao–Bordeaux Workshop on PDEs**, BCAM, Bilbao, Spain
Feb–Apr 2024	**BCAM SO Course: Introduction to the Mathematical Theory of Incompressible Fluids** *Lecturers:* F. Fanelli and R. Lucà – BCAM, Bilbao, Spain †

† Funding received to attend.

Seminars and Reading Groups

Nov 2024–present	**PhD Seminar & Courses**, IMUS, University of Seville
Apr 2024–Jun 2026	**Fluid Dynamics Reading Group**, US–UGR
Oct 2023–present	**Fluid Conversations Seminar Series**, IMUS, University of Seville

Organization and Service

Dec 2025–present	**Member of the IMUS Council** (PhD student representative), University of Seville
Nov 2025–present	**Member (PhD student representative)**, Internal Quality Assurance Committee (CGICPD), PhD Programme in Mathematics, University of Seville
Oct 2024–Jul 2025	**Co-organizer**, PHD Seminar, IMUS, University of Seville
Feb 2025	**Student participant**, stakeholder hearing for the renewal of accreditation of the PhD Programme in Mathematics, University of Seville
Jul 2024	**Volunteer**, 9th European Congress of Mathematics (ECM), Seville

Science Outreach

Feb 2026	**Workshop: Statistics with Sugus**, IES Pésula
Feb 2026	**Workshop: Four Color Theorem**, Colegio Aljarafe S.C.A.
Jun 2025	**Talk at the closing session** of *Una científica en tu cole andaluz*
Mar 2025	**Workshop: Four Color Theorem**, CEI Julio César
Feb 2025	**Workshop: Building Mosaics with Mathematics**, CEIP Los Eucaliptos
Jan 2025	**Talk: Decipher Fluids, Win a Million**, for IES Triana students visiting IMUS
Feb 2024	**Talk: Women and Mathematics**, IES San Pablo

References

Francisco Gancedo – PhD advisor, University of Seville, Spain fgancedo@us.es
Eduardo García-Juárez – PhD advisor, University of Seville, Spain egarcia12@us.es
Francisco Mengual – Collaborator, University of Seville, Spain fmengual@us.es
Raphaël Danchin – Research stay host, Université Paris-Est Créteil (UPEC), France danchin@u-pec.fr